Indian Heart Journal 76 (2024) S86–S89

Contents lists available at ScienceDirect

Indian Heart Journal

journal homepage: www.elsevier.com/locate/ihj

Review Article

Dyslipidemia and peripheral arterial disease

Ajay Yadav [a,*], Vivek Sawant [a], Varinder Singh Bedi [a], Kanupriya Yadav [b]

[a] Institutions: Sir Ganga Ram Hospital, India
[b] Mahajan Imaging and Labs, India

ARTICLE INFO

Keywords:
Dyslipidaemia
Peripheral arterial disease
Ezetimibe
Polyvascular disease
LDL-C goal of <55 mg/dl

ABSTRACT

Peripheral arterial disease (PAD) affects 12 % of adult population and is increasing globally and in India. Peripheral arterial disease when associated with atherosclerosis in two or more other arterial beds such as coronary artery disease (CAD), mesenteric/renal artery and cerebrovascular disease (CVD), is known as polyvascular disease. The Reduction of Atherothrombosis for Continued Health (REACH) registry reported that 1 out of 6 patients had multi-vascular bed involvement. Progression of PAD to critical limb ischaemia (CLI) is seen in 1 % of affected patients per year, but patients who progress to CLI may have a 10- to 15-fold increased risk of cardiovascular death.

The 2019 ECS/EAS guidelines for the management of dyslipidaemias have suggested that for primary or secondary prevention in very high risk, patients should follow a therapeutic regimen that achieves >50 % LDL-C reduction from baseline and an LDL-C goal of <55 mg/dl. High Intensity Statin is mainstay of treatment but optimal management is inadequate. Statin treatment reduces all-cause mortality by 39 %, CV death by 41 %, CV outcomes by 34 %, ischaemic stroke by 28 %, acute limb ischaemia by 30 % and amputations by 35 %. Ezetimibe when added to statins in IMPROVE-IT trial, showed significant reduction of MACE. PCSK9 inhibitor (FOURIER TRIAL) showed reduction in primary end point in PAD vs Non PAD patients (3.5 % vs 1.6 %).

There is a critical need for an Indian multi-disciplinary task force for research on the direct impact of lipid-lowering agents on limb salvage rates and major limb-related events in PAD patients.

1. Introduction

Peripheral arterial disease (PAD), characterised by occlusion or narrowing of the arteries supplying the lower extremities, has recently been estimated to affect approximately 240 million people worldwide, a large section (about 70 %) of whom reside in low- or middle-income countries.[1] Peripheral arterial disease reportedly affects up to 12 % of the adult population in Western countries, and the incidence increases with age to as high as 29 % for those aged >70 years.[2] Peripheral arterial disease prevalence data for India is sparse and one study from Kerala reported a rate of nearly 27 %.[3]

Peripheral arterial disease is a manifestation of atherosclerosis and is associated with an elevated risk of myocardial infarction and cardiovascular deaths. Peripheral arterial disease when associated with atherosclerosis in two or more other arterial beds such as coronary artery disease (CAD), mesenteric/renal artery and cerebrovascular disease (CVD), is known as polyvascular disease.[4] The Reduction of Atherothrombosis for Continued Health (REACH) registry reported that 1 out of 6 patients had multi-vascular bed involvement. Subsequent studies have largely verified this prevalence of PAD in the population of patients with stable CAD.[5–7] The REACH study demonstrated that major adverse cardiac event rates are significantly higher (nearly three-fold cumulative increase) among patients with PAD compared with those with single vascular bed disease or those with only risk factors for atherosclerotic disease. Recent analyses of randomised trials geared towards secondary prevention in patients with CAD have shown up to 50 % of major adverse cardiac events (MACE) in patients with PAD.[8,9]

Progression of PAD to critical limb ischaemia (CLI) is seen in 1 % of affected patients per year, but patients who progress to CLI may have a 10- to 15-fold increased risk of cardiovascular death.[10]

1.1. Dyslipidaemia

Dyslipidaemia (especially high LDL cholesterol) predicts the risk of cardiovascular events, and PAD patients are more likely to have higher rates of hyperlipidaemia and other risk factors.

* Corresponding author.
E-mail address: ajay.vascular@gmail.com (A. Yadav).

https://doi.org/10.1016/j.ihj.2024.01.010
Received 14 October 2023; Received in revised form 12 December 2023; Accepted 12 January 2024
Available online 13 January 2024
0019-4832/© 2024 Cardiological Society of India. Published by Elsevier, a division of RELX India, Pvt. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Treatment of dyslipidaemia indirectly reduces cardiovascular mortality and morbidity in patients with PAD and should include lipid-lowering therapies. The 2019 ECS/EAS guidelines for the management of dyslipidaemias have suggested that for primary or secondary prevention in very high risk, patients should follow a therapeutic regimen that achieves >50 % LDL-C reduction from baseline and an LDL-C goal of <55 mg/dl.[11]

However, despite these recommendations as well as numerous studies showing the benefit of lipid-lowering treatment in PAD patients, optimal medical management remains inadequate.[8,12,13] The ALLIANCE study noted less than half of patients presenting with PAD with ACS received the guideline-recommended medications.[14] Zhang et al reported that asymptomatic patients with PAD as detected by ankle-brachial index (ABI) and transcranial doppler screening had higher rates of MACE at 4 years.[13] Screening population with PAD for dyslipidaemia, which is a modifiable risk factor, can have a significant impact.

1.2. Management of dyslipidaemia

Diet is the first important and mandatory step for lowering serum cholesterol, along with medical therapy to achieve recommended LDL-C goals. A systematic review of 18 trials including 10,000 patients, with cholesterol levels ranging from normal to elevated, reported that lipid-lowering therapy in people affected by lower limb atherosclerosis was associated with a 20 % reduction in total CV events, together with a non-significant 14 % reduction in all-cause mortality.[15]

1.3. Role of statins

A meta-analysis including 18,686 patients with diabetes mellitus (DM) demonstrated that a statin-induced reduction of LDL-C by 1.0 mmol/l (40 mg/dl) was associated with a 9 % reduction in all-cause mortality and a 21 % reduction in the incidence of major CV events. In a meta-analysis of RCTs enrolling >90,000 patients on statin therapy, which considered mainly the extent of LDL-C reduction, showed a 21 % reduction in the incidence of all strokes in different populations.[16]In the Heart Protection Study, the need for non-coronary revascularisation was reduced by 16 % with statin therapy. Further, statin treatment has been reported to reduce all-cause mortality by 39 %, CV death by 41 %, CV outcomes by 34 % and ischaemic stroke by 28 %, but also reduced acute limb ischaemia by 30 % and amputations by 35 %.[16]

In non-cardiac vascular procedures, perioperative statin therapy significantly reduced the incidence of fatal and non-fatal cardiac events, congestive heart failure, stroke, vascular complications, length of stay, and the overall mortality rate.[17–19] In contrast, perioperative discontinuation of statin therapy was associated with increased cardiac events.[20]

1.4. Impact on limb salvage

Patients who were receiving statin therapy at the time of endovascular intervention for CLI had improved patency, limb salvage, and overall survival.[21] Statin therapy is associated with improved patency after infra-inguinal bypass graft surgery and improved outcomes after endovascular aortic aneurysm repair, carotid artery stenting.[22]

The Scandinavian Simvastatin Survival Study revealed a 38 % reduction in new and worsening intermittent claudication in addition to the already proven cardiovascular benefits in patients with CAD.[23,24] Statins also improved ABI, walking distance, and pain-free walking distance. The HPS, a trial of simvastatin, 40 mg/day, versus antioxidant vitamins or placebo in 20,536 participants at high risk for a CHD event[24] showed a 20 % reduction in non-coronary revascularisation, including a decrease in the number of carotid endarterectomies. The results of HPS which was a landmark trial supports the treatment of all patients with PAD with a statin agent for lipid-lowering therapy.[25]

The Prospective Study of Pravastatin in the Elderly at Risk (PROSPER) wherein high-risk elderly population (5800 patients with a history of pre-existing vascular disease, or at increased risk) was studied were randomised to 40 mg/day of pravastatin or placebo showed that need for peripheral vascular intervention requirement was less in treatment group compared to placebo group (HR 0.78, $p = 0.25$).[23]

1.5. Role of ezetimibe

IMPROVE-IT trial[26] reported that in patients with PAD, ezetimibe had largest absolute risk reduction on MACE, possibly due to the higher risk of this subgroup.[8,9,27] Ezetimibe has been proved to lower LDL-C when added to statins, which was assessed in patients with PAD.[28] Evolution of atherosclerotic plaques in the superficial femoral artery was studied by using magnetic resonance imaging in patients with PAD, treated with statin or statin plus ezetimibe.[21] Statin initiation, with/-without ezetimibe, proved to stop the progression of atherosclerosis. Adding ezetimibe to patients previously treated only with statin surprisingly led to a progression in peripheral atherosclerosis, despite the 22 % decrease in LDL-C. These results correlate with those of a different study which compared the effects of niacin added to statin to ezetimibe added to statin therapy. Hence, role of ezetimibe in reducing risk of CLI in patients with PAD is not clear.

1.6. Role of proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor

In the Further Cardiovascular Outcomes Research with PCSK9 Inhibition in Subjects with Elevated Risk (FOURIER) trial,[29] patients with atherosclerotic disease on statin therapy were randomly assigned to a fixed dose of evolocumab or placebo. The PCSK9 inhibition led to a 47 % reduction in LDL-cholesterol despite on statin therapy. Evolocumab significantly reduced the MACE by 21 %, as well as the risk of MALE by 42 %.[30] Thus the FOURIER was the first randomized trial to demonstrate that intensive LDL-C lowering was safe and effective in decreasing the risk of acute limb ischaemia.[16] Similarly, in Alirucumab and cardiovascular outcome after acute coronary syndrome (ODYSSEY) trial wherein primary endpoint was composite death with a median follow up of 2.8 years. In this study, patients on high intensity statins were randomized to receive alirocumab or placebo with dose adjustment every 2 weeks. Results of ODYSSEY trial showed that primary event occurred in 9.5 % vs 11.1 % in alirocumab and placebo group respectively suggesting that risk of recurrent ischaemia is lower with aggressive statin therapy. A subgroup analysis of patients with history of PAD or VTE within ODYSSEY OUTCOMES on comparing with placebo demonstrated lower PAD related events in 2.5 % of patients.[31] Larger population studies are required in order to assess their possible role in PAD.

2. Role of fibrates

Both the Fenofibrate Intervention and Event Lowering in Diabetes (FIELD)[32] and ACCORD[33] studies demonstrated that administration of fenofibrate on top of statins significantly reduced microvascular complications and carotid revascularisations. The fibrates, gemfibrozil and fenofibrate are peroxisome proliferator receptor-α agonists used for treating hypertriglyceridemia and to increase HDL-C.[34] Fenofibrate reduced the risk of minor amputations without known large-vessel disease through non-lipid mechanisms.[31]

Triglyceride lowering with Pemafibrate to Reduce Cardiovascular Risk (PROMINENT trial) showed that among patients with type 2 diabetes, mild-to-moderate hypertriglyceridemia, and low HDL and LDL cholesterol levels, the incidence of cardiovascular events was not lower among those who received pemafibrate than among those who received placebo, although pemafibrate lowered triglyceride, VLDL cholesterol, remnant cholesterol, and apolipoprotein C-III levels.

3. Role of icosapent ethyl

The Reduction of Cardiovascular Events with Icosapent Ethyl-Intervention Trial (REDUCE-IT), which enrolled more than 8000 patients for a time period of nearly 5 years showed that the patients receiving icosapent ethyl (2 gm BD with food) were 25 % less likely than placebo patients to experience cardiovascular death, nonfatal MI, stroke, coronary revascularization or unstable angina.[35] Prescription omega-3 fatty acids (eicosapentanoic acid – EPA and/or docosahexaenoic acid-DHA) are indicated to treat severe hypertriglyceridemia (TG levels of 500 mg/dL or higher).[36]

4. Role of bempedoic acid

Bempedoic acid and cardiovascular outcomes in statin intolerant patients trial (CLEAR outcomes) published in 2023 showed that among statin-intolerant patients, treatment with bempedoic acid was associated with a lower risk of major adverse cardiovascular events (death from cardiovascular causes, nonfatal myocardial infarction, nonfatal stroke, or coronary revascularization).[37]

5. Role of inclisiran

Two Phase 3 Trials of Inclisiran in Patients with Elevated LDL Cholesterol (ORION 10 & ORION 11 trials) showed that reductions in LDL cholesterol levels of approximately 50 % were obtained with inclisiran, administered subcutaneously every 6 months. More injection-site adverse events occurred with inclisiran than with placebo.[38]

6. Conclusion

There is a critical need for Indian research data on the direct impact of lipid-lowering agents on limb salvage rates and major limb-related events in PAD patients. Adoption of specific risk assessment scores for PAD in dyslipidaemia that are tailored to the Indian population is urgent. A multidisciplinary task force is also required to develop guidelines for managing dyslipidaemia in Indians in order to reduce cardiovascular risk.

Anti-lipid agent	Trials
Statins	**Heart Protection Study**
	Scandinavian Simvastatin Survival Study
	PROSPER
Ezetimibe	**IMPROVE-IT**
PCSK9	**FOURIER**
	ODYSSEY
Fibrates	**FIELD**
	ACCORD
	PROMINENT
Icosapent Ethyl	**REDUCE-IT**
Bempedoic Acid	**CLEAR Outcomes**
Inclisiran	**ORION 10 & ORION 11**

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Golledge J, Fernando ME, Armstrong DG. Current management of peripheral artery disease: focus on pharmacotherapy. *Drugs*. 2022;82(11):1165–1177, 10.1007. s40265-022-01755-7. Epub 2022 Aug 12.
2. Society E, France MJC, Backer GG De, et al. ESC Scientific Document Group. 2019 ESC/EAS Guidelines for the management of dyslipidaemias : lipid modification to reduce cardiovascular risk. *Eur Heart J*. 2020:111–188.
3. Krishnan MN, Geevar Z, Mohanan PP, Venugopal K, Devika S. Prevalence of peripheral artery disease and risk factors in the elderly: a community based cross-sectional study from northern Kerala, India. *Indian Heart J*. 2018;70(6):808–815. https://doi.org/10.1016/j.ihj.2017.11.001. Epub 2017 Nov 4.
4. Gutierrez JA, Aday AW, Patel MR, Jones WS. Polyvascular disease: reappraisal of the current clinical landscape. *Circ Cardiovasc Interv*. 2019;12, e007385.
5. *The World Health Report 2002: Reducing Risks, Promoting Healthy Lives*. Geneva: World Health Organization; 2002.
6. Mas JL, Pinel JF, Priollet P, et al. REACH: international prospective observational registry in patients at risk of atherothrombotic events Results for the French arm at baseline and one year. *Arch Cardiovasc Dis*. 2008;101(2):81–88.
7. Bhatt DL, Steg PG, Ohman EM, et al. International prevalence recognition, and treatment of cardiovascular risk factors in outpatients with Atherothrombosis. *JAMA*. 2006;295:180–189.
8. Bonaca MP, Bhatt DL, Storey RF, et al. Ticagrelor for prevention of ischemic events after myocardial infarction in patients with peripheral artery disease. *J Am Coll Cardiol*. 2016;67:2719–2728.
9. Franzone A, Piccolo R, Gargiulo G, et al. Prolonged vs short duration of dual antiplatelet therapy after percutaneous coronary intervention in patients with or without peripheral arterial disease: a subgroup analysis of the PRODIGY randomized clinical trial. *JAMA Cardiol*. 2016;1:795–803.
10. Criqui MH, Langer RD, Fronek A, et al. Mortality over a period of 10 years in patients with peripheral arterial disease. *N Engl J Med*. 1992;326:381–386.
11. Catapano AL, Graham I, De Backer G, , et alESC Scientific Document Group. ESC/EAS Guidelines for management of dyslipidaemias. *Eur Heart J*. 2016;37:2999–3058, 2016.
12. Vlajinac HD, Marinkovic JM, Maksimovic MZ, et al. The prevalence of polyvascular disease in patients with carotid artery disease and peripheral artery disease. *Kardiol Pol*. 2019;77:926–934.
13. Zhang Q, Wang A, Zhang S, et al. Asymptomatic polyvascular disease and the risks of cardiovascular events and all-cause death. *Atherosclerosis*. 2017;262:1–7.
14. Meizels A, Messika Zeitoun D, et al. Impact of polyvascular disease on baseline characteristics, management and mortality in acute myocardial infarction. *Arch Cardiovasc Dis*. 2010;103:207–214.
15. Aung PP, Maxwell HG, Jepson RG, et al. Lipid-lowering for peripheral arterial disease of the lower limb. *Cochrane Database Syst*. 2007;4, CD000123.
16. Belch JJF, Brodmann M, Baumgartner I, Binder CJ, et al. Lipid-lowering and anti-thrombotic therapy in patients with peripheral arterial disease: European atherosclerosis society/European society of vascular medicine joint statement. *Atherosclerosis*. 2021;338:55–63. https://doi.org/10.1016/j.atherosclerosis.2021.09.022. Epub 2021 Nov 8.
17. Kertai MD, Boersma E, Westerhout CM, et al. A combination of statins and beta-blockers is associated with a reduction in the incidence of perioperative mortality in nonfatal myocardial infarction in patients undergoing abdominal aortic aneurysm surgery. *Eur J Vasc Endovasc Surg*. 2004;28:343–352.
18. Durazzo AE, Machado FS, Ikeoka DT, et al. Reduction in cardiovascular events after vascular surgery with atorvastatin: a randomized trial. *J Vasc Surg*. 2004;39:967–976.
19. Schouten O, Boersma E, Hoeks SE, et al. Dutch echocardiographic cardiac risk evaluation applying stress echocardiography study group. Fluvastatin and perioperative events in patients undergoing vascular surgery. *N Engl J Med*. 2009;361:980–989.
20. Schouten O, Hoeks SE, Welten GM, et al. Effect of statin withdrawal on frequency of cardiac events after vascular surgery. *Am J Cardiol*. 2007;100:316–320.
21. Youssef F, Seifalian AM, Jagroop IA, et al. The early effect of lipid lowering treatment on carotid and femoral intima media thickness (IMT). *Eur J Vasc Endovasc Surg*. 2002;23(4):358–364.
22. Ohman EM, Bhatt DL, Steg PG, et al. The Reduction of Atherothrombosis for Continued Health (REACH) Registry: an international, prospective, observational investigation in subjects at risk for atherothrombotic events-study design. *J Am Coll Cardiol*. 2006;151:e1–e10.
23. Shepherd J, Blauw GJ, Murphy MB, et al. Pravastatin in elderly individuals at risk of vascular disease (PROSPER): a randomised controlled trial. *Lancet*. 2002;360(9346):1623–1630.
24. Group HPSC. Randomized trial of the effects of cholesterol-lowering with simvastatin on peripheral vascular and other major vascular outcomes in people with peripheral arterial disease and other high-risk conditions. *J Vasc Surg*. 2007;45(4):645–654.
25. Collins R, Armitage J, Parish S, et al. Heart protection study cholesterol lowering, collaborative group. MRC/BHF heart protection study of placebo controlled, with simvastatin in 5963 people with diabetes: a randomised trial. *Lancet*. 2003;361:2005–2016.
26. Bonaca MP, Antonio Gutierrez J, Cannon C, et al. Polyvascular disease, type 2 diabetes, and long-term vascular risk: a secondary analysis of the IMPROVE-IT trial. *Lancet Diabetes Endocrinol*. 2018;6:934–943.
27. Bonaca MP. Polyvascular disease and risk: when two is not better than one. *Vasc Med*. 2018;23:531–553.
28. West AM, Anderson JD, Meyer CH, et al. The effect of ezetimibe on peripheral arterial atherosclerosis depends upon statin use at baseline. *Atherosclerosis*. 2011;218:156–162.
29. Sabatine MS, Giugliano RP, Keech AC, et al. Evolocumab and clinical outcomes in patients with cardiovascular disease. *N Engl J Med*. 2017;376(18):1713–1722.
30. Bonaca MP, Nault P, Giugliano RP, et al. Low-density lipoprotein cholesterol lowering with evolocumab and outcomes in patients with peripheral artery disease: insights from the FOURIER trial (Further Cardiovascular Outcomes Research with PCSK9 Inhibition in Subjects with Elevated Risk). *Circulation*. 2018;137:338–350.
31. Schwartz GG, Steg PG, Szarek M, et al. Peripheral artery disease and venous thromboembolic events after acute coronary syndrome: role of lipoprotein(a) and

modifica- tion by alirocumab: prespecified analysis of the ODYSSEY OUTCOMES randomized clinical trial. *Circulation*. 2020;141:1608–1617.
32. Field T. Effects of long-term fenofibrate therapy on cardiovascular events in 9795 people with type 2 diabetes mellitus (the FIELD study): randomised controlled trial. *Lancet*. 2005;366(9500):1849–1861.
33. Elam M, Lovato L, Ginsberg H. The ACCORD-Lipid study: implications for treatment of dyslipidemia in Type 2 diabetes mellitus. *Clin Lipidol*. 2011;6(1):9–20.
34. Tendera M, Aboyans VBM, Baumgartner I, et al. ESC Committee for Practice Guidelines. ESC Guidelines on the diagnosis and treatment of peripheral artery diseases: document covering atherosclerotic disease of extracranial carotid and vertebral, mesenteric, renal, upper and lower extremity arteries. *Eur Heart J*. 2011: 2851–2906.
35. Bhatt DL, Steg PG, Brinton EA, Jacobson TA, Miller M, et al. REDUCE-IT investigators. Rationale and design of REDUCE-IT: reduction of cardiovascular events with icosapent ethyl-intervention trial. *Clin Cardiol*. 2017 Mar;40(3): 138–148.
36. Hilleman DE, Wiggins BS, Bottorff MB. Critical differences between dietary supplement and prescription omega-3 fatty acids: a narrative review. *Adv Ther*. 2020;37:656–670.
37. Nissen SE, Lincoff AM, Brennan D, Ray KK, Mason D, et al. CLEAR outcomes investigators. Bempedoic acid and cardiovascular outcomes in statin-intolerant patients. *N Engl J Med*. 2023 Apr 13;388(15):1353–1364.
38. Ray KK, Wright RS, Kallend D, et al. ORION-10 and ORION-11 investigators. Two Phase 3 trials of inclisiran in patients with elevated LDL cholesterol. *N Engl J Med*. 2020 Apr 16;382(16):1507–1519.